

48436/32309

SBA Report

You will have 60 mins to work on any three out four questions. Each question is worth 10 pts.

You are a forensics investigator working in a team investigating a suspect.

Q1 Week pcap Questions (Replace this with yours)

a) What filter did you use? DNS

Packet list	Filter	View	Case sensitive	String	Offensive	Find	Go
No.	Time	Source	Destination	Protocol	Length	Info	
927	12.220048	10.0.0.138	10.0.0.14	DNS	102	Standard query response 0xc10 A services.disconnect.me CNAME services.disconnect.me.herokuks.com A 35.172.84.218 A 3.212.40.153 A 54.164.152.140 A 54.85.41.146 A 107.23.162.152 A 54.16...	
940	12.223908	10.0.0.138	10.0.0.14	DNS	102	Standard query response 0xc216 A encrypted-tb0.gstatic.com A 216.58.199.78	
957	12.243187	10.0.0.138	10.0.0.14	DNS	136	Standard query response 0xc5cf A adservice.google.com CNAME pagead44.l.doubleclick.net A 172.217.167.80	
1276	14.349397	10.0.0.138	10.0.0.14	DNS	114	Standard query response 0xc162 A www.officeworks.com.au A 52.62.251.32 A 13.239.51.109	
1429	15.149927	10.0.0.138	10.0.0.14	DNS	117	Standard query response 0xc040 A officeworks.tt.omtrdc.net A 1304.171.81 A 13.211.113.30	
1449	15.216822	10.0.0.138	10.0.0.14	DNS	167	Standard query response 0xc040 A apps.bazaarvoice.com CNAME d3rpaigr3c5p5n.cloudfront.net A 99.86.212.26 A 99.86.212.43 A 99.86.212.111 A 99.86.212.62	
1478	15.351089	10.0.0.138	10.0.0.14	DNS	302	Standard query response 0xc880 A static.hotjar.com CNAME static.hotjar.com.cdn.cloudflare.net A 103.43.119.214 A 110.232.118.141 A 103.4...	
1541	15.455365	10.0.0.138	10.0.0.14	DNS	116	Standard query response 0xc87c A ipapi.co A 104.26.9.44 A 104.26.8.44 A 172.67.69.226	
1584	15.644920	10.0.0.138	10.0.0.14	DNS	287	Standard query response 0xc0ff4 A dpm.dendex.net CNAME gslb-2.dendex.net CNAME edge-aspe2.dendex.net CNAME dcs-edge-aspe2-246104331.ap-southeast-2.elb.amazonaws.com A 3.24.64.160 A 3.104...	
1597	15.662577	10.0.0.138	10.0.0.14	DNS	179	Standard query response 0xc780 A assets.addednet.com CNAME cn-assets.addednet.com.edgekey.net CNAME e7800.dig.akamaiedge.net A 104.116.102.105	
1619	15.697151	10.0.0.138	10.0.0.14	DNS	141	Standard query response 0xc0e2 A s.yimg.com CNAME edge.gyrfp.b.yahoodns.net A 115.178.9.19 A 115.178.9.18	
1680	15.951295	10.0.0.138	10.0.0.14	DNS	295	Standard query response 0xc295 A officeworks.dendex.net CNAME gslb-2.dendex.net CNAME edge-aspe2.dendex.net CNAME dcs-edge-aspe2-246104331.ap-southeast-2.elb.amazonaws.com A 3.24.64.160	
1687	15.954530	10.0.0.138	10.0.0.14	DNS	133	Standard query response 0xc07a A officeworks.sc.omtrdc.net A 52.62.21.192 A 54.66.152.159 A 13.238.77.77	
1696	15.955390	10.0.0.138	10.0.0.14	DNS	134	Standard query response 0xc092 A cn.everesttech.net CNAME cn.everesttech.net.akadns.net A 66.117.25.58	
1699	15.962394	10.0.0.138	10.0.0.14	DNS	124	Standard query response 0xc731 A 4308958.fls.doubleclick.net CNAME dart.1.doubleclick.net A 172.217.167.70	
1814	16.109414	10.0.0.138	10.0.0.14	DNS	115	Standard query response 0xc0d6 A ad.doubleclick.net CNAME dart.1.doubleclick.net A 172.217.167.70	
1826	16.132942	10.0.0.138	10.0.0.14	DNS	124	Standard query response 0xc09f A 6254122.fls.doubleclick.net CNAME dart.1.doubleclick.net A 172.217.167.70	
1850	16.170408	10.0.0.138	10.0.0.14	DNS	140	Standard query response 0xc027 A www.bloggerads.com CNAME mix.linkedin.com CNAME pop-es3.mix.linkedin.com A 144.2.5.5	
1856	16.183007	10.0.0.138	10.0.0.14	DNS	128	Standard query response 0xc743 A googleads.g.doubleclick.net CNAME pagead44.l.doubleclick.net A 172.217.167.90	
1878	16.200431	10.0.0.138	10.0.0.14	DNS	176	Standard query response 0xc087 A bat.bing.com CNAME bat-bing.com-a-0001.a-msedge.net CNAME dual-a-0001.a-msedge.net A 13.107.21.200 A 204.79.197.200	
2002	16.247028	10.0.0.138	10.0.0.14	DNS	180	Standard query response 0xc087 A bat.bing.com CNAME bat-bing.com-a-0001.a-msedge.net CNAME dual-a-0001.a-msedge.net A 13.107.21.200 A 204.79.197.200	
2029	16.261818	10.0.0.138	10.0.0.14	DNS	135	Standard query response 0xc79c A sync.mathtag.com CNAME pixel-origin.mathtag.com A 183.229.206.39 A 183.229.206.182	
2062	16.288378	10.0.0.138	10.0.0.14	DNS	146	Standard query response 0xc79c A sync.mathtag.com CNAME pixel-origin.mathtag.com A 183.229.206.182 A 183.229.206.39	
2078	16.314442	10.0.0.138	10.0.0.14	DNS	201	Standard query response 0xc63c A in.hotjar.com A 34.240.31.203 A 52.214.117.39 A 63.32.141.164 A 18.202.211.106 A 34.246.206.139 A 54.171.1.253 A 52.49.171.198 A 52.51.24.70	
Frame 1276: 114 bytes on wire (912 bits), 114 bytes captured (912 bits) on interface 'Device\NPF_{61834A95-AB7A-4F29-9408-F536A2122C28}'							
Ethernet II, Src: Sagecom-cd03:f6 (f6:ab05:cd03:f6), Dst: ASUSLAC-02:4e:23 (9c:5c:be02:4e:23)							
Internet Protocol Version 4, Src: 10.0.0.138, Dst: 10.0.0.14							
User Datagram Protocol, Src Port: 53, Dst Port: 68513							
Domain Name System (response)							
Transaction ID: 0xc162							
Flags: 0x0180 Standard query response, No error							
Questions: 1							
Answer RRs: 2							
Authority RRs: 0							
Additional RRs: 0							
Queries							
www.officeworks.com.au: type A, class IN							
Answers							
www.officeworks.com.au: type A, class IN, addr 52.62.251.32							
www.officeworks.com.au: type A, class IN, addr 13.239.51.109							
[Request In: 1455]							
[Time: 0.021479000 seconds]							

b) What is the purpose of Bazaarvoice? Bazaarvoice is basically an analytics reviews platform.

1429	15.149927	10.0.0.138	10.0.0.14	DNS	117	Standard query response 0xc040 A officeworks.tt.omtrdc.net A 1304.117.81 A 13.211.113.30
1449	15.216822	10.0.0.138	10.0.0.14	DNS	167	Standard query response 0xc040 A apps.bazaarvoice.com CNAME d3rpaigr3c5p5n.cloudfront.net A 99.86.212.26 A 99.86.212.43 A 99.86.212.111 A 99.86.212.62
1478	15.351089	10.0.0.138	10.0.0.14	DNS	302	Standard query response 0xc880 A static.hotjar.com CNAME static.hotjar.com.cdn.cloudflare.net A 103.43.119.214 A 110.232.118.141 A 103.4...
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1584	15.644920	10.0.0.138	10.0.0.14	DNS	287	Standard query response 0xc0ff4 A dpm.dendex.net CNAME gslb-2.dendex.net CNAME edge-aspe2.dendex.net CNAME dcs-edge-aspe2-246104331.ap-southeast-2.elb.amazonaws.com A 3.24.64.160 A 3.104...
1597	15.662577	10.0.0.138	10.0.0.14	DNS	179	Standard query response 0xc780 A assets.addednet.com CNAME cn-assets.addednet.com.edgekey.net CNAME e7800.dig.akamaiedge.net A 104.116.102.105
1619	15.697151	10.0.0.138	10.0.0.14	DNS	141	Standard query response 0xc0e2 A s.yimg.com CNAME edge.gyrfp.b.yahoodns.net A 115.178.9.19 A 115.178.9.18
1680	15.951295	10.0.0.138	10.0.0.14	DNS	295	Standard query response 0xc295 A officeworks.dendex.net CNAME gslb-2.dendex.net CNAME edge-aspe2.dendex.net CNAME dcs-edge-aspe2-246104331.ap-southeast-2.elb.amazonaws.com A 3.24.64.160
1687	15.954530	10.0.0.138	10.0.0.14	DNS	133	Standard query response 0xc07a A officeworks.sc.omtrdc.net A 52.62.21.192 A 54.66.152.159 A 13.238.77.77
1696	15.955390	10.0.0.138	10.0.0.14	DNS	134	Standard query response 0xc092 A cn.everesttech.net CNAME cn.everesttech.net.akadns.net A 66.117.25.58
1814	16.109414	10.0.0.138	10.0.0.14	DNS	115	Standard query response 0xc0d6 A ad.doubleclick.net CNAME dart.1.doubleclick.net A 172.217.167.70
1826	16.132942	10.0.0.138	10.0.0.14	DNS	124	Standard query response 0xc09f A 6254122.fls.doubleclick.net CNAME dart.1.doubleclick.net A 172.217.167.70
1850	16.170408	10.0.0.138	10.0.0.14	DNS	140	Standard query response 0xc027 A www.bloggerads.com CNAME mix.linkedin.com CNAME pop-es3.mix.linkedin.com A 144.2.5.5
1856	16.183007	10.0.0.138	10.0.0.14	DNS	128	Standard query response 0xc743 A googleads.g.doubleclick.net CNAME pagead44.l.doubleclick.net A 172.217.167.90
1878	16.200431	10.0.0.138	10.0.0.14	DNS	176	Standard query response 0xc087 A bat.bing.com CNAME bat-bing.com-a-0001.a-msedge.net CNAME dual-a-0001.a-msedge.net A 13.107.21.200 A 204.79.197.200
2002	16.247028	10.0.0.138	10.0.0.14	DNS	180	Standard query response 0xc087 A bat.bing.com CNAME bat-bing.com-a-0001.a-msedge.net CNAME dual-a-0001.a-msedge.net A 13.107.21.200 A 204.79.197.200
2029	16.261818	10.0.0.138	10.0.0.14	DNS	135	Standard query response 0xc79c A sync.mathtag.com CNAME pixel-origin.mathtag.com A 183.229.206.39 A 183.229.206.182
2062	16.288378	10.0.0.138	10.0.0.14	DNS	146	Standard query response 0xc79c A sync.mathtag.com CNAME pixel-origin.mathtag.com A 183.229.206.182 A 183.229.206.39
2078	16.314442	10.0.0.138	10.0.0.14	DNS	201	Standard query response 0xc63c A in.hotjar.com A 34.240.31.203 A 52.214.117.39 A 63.32.141.164 A 18.202.211.106 A 34.246.206.139 A 54.171.1.253 A 52.49.171.198 A 52.51.24.70
Frame 1409: 187 bytes on wire (1496 bits), 187 bytes captured (1496 bits) on interface 'Device\NPF_{61834A95-AB7A-4F29-9408-F536A2122C28}'						
Ethernet II, Src: Sagecom-cd03:f6 (f6:ab05:cd03:f6), Dst: ASUSLAC-02:4e:23 (9c:5c:be02:4e:23)						
Internet Protocol Version 4, Src: 10.0.0.138, Dst: 10.0.0.14						
User Datagram Protocol, Src Port: 53, Dst Port: 59947						
Domain Name System (response)						
Transaction ID: 0xc0d6						
Flags: 0x0180 Standard query response, No error						
Questions: 1						
Answer RRs: 5						
Authority RRs: 0						
Additional RRs: 0						
Queries						
apps.bazaarvoice.com: type A, class IN						
Answers						
apps.bazaarvoice.com: type CNAME, class IN, cname d3rpaigr3c5p5n.cloudfront.net						
d3rpaigr3c5p5n.cloudfront.net: type A, class IN, addr 99.86.212.26						
d3rpaigr3c5p5n.cloudfront.net: type A, class IN, addr 99.86.212.43						
d3rpaigr3c5p5n.cloudfront.net: type A, class IN, addr 99.86.212.111						
d3rpaigr3c5p5n.cloudfront.net: type A, class IN, addr 99.86.212.62						
Request In: 1407						
[Time: 0.014830000 seconds]						

What website is hosting bazaarvoice? d3rpaigr3c5p5n.cloudfront.net

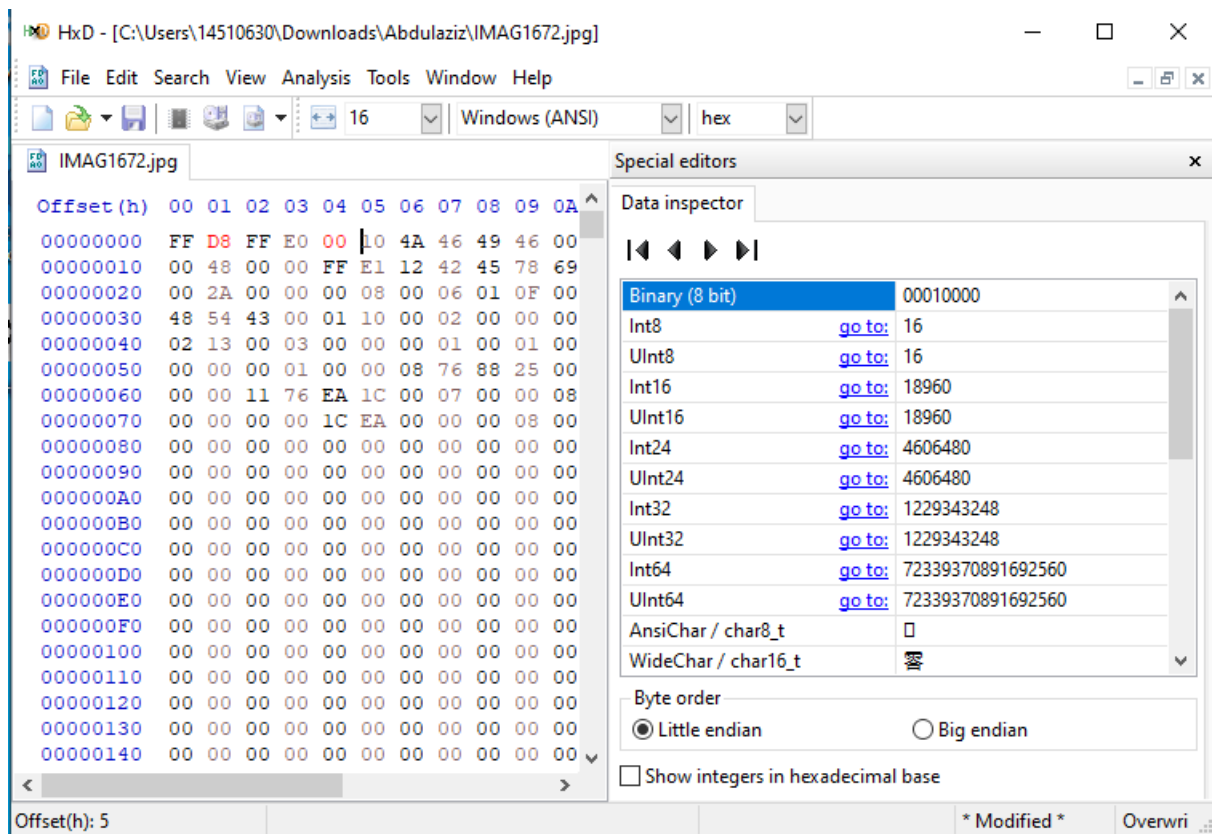
c)

SBA Report

What website is hosting demdex? dcs-edge-apse2-246104331.ap-southeast-2.elb.amazonaws.com

a) Use the xxd command to check what is wrong with the HEX header.

Use HXD or similar to Repair the HEX header so the image displays. Take a **screenshot** of the repaired first header line.



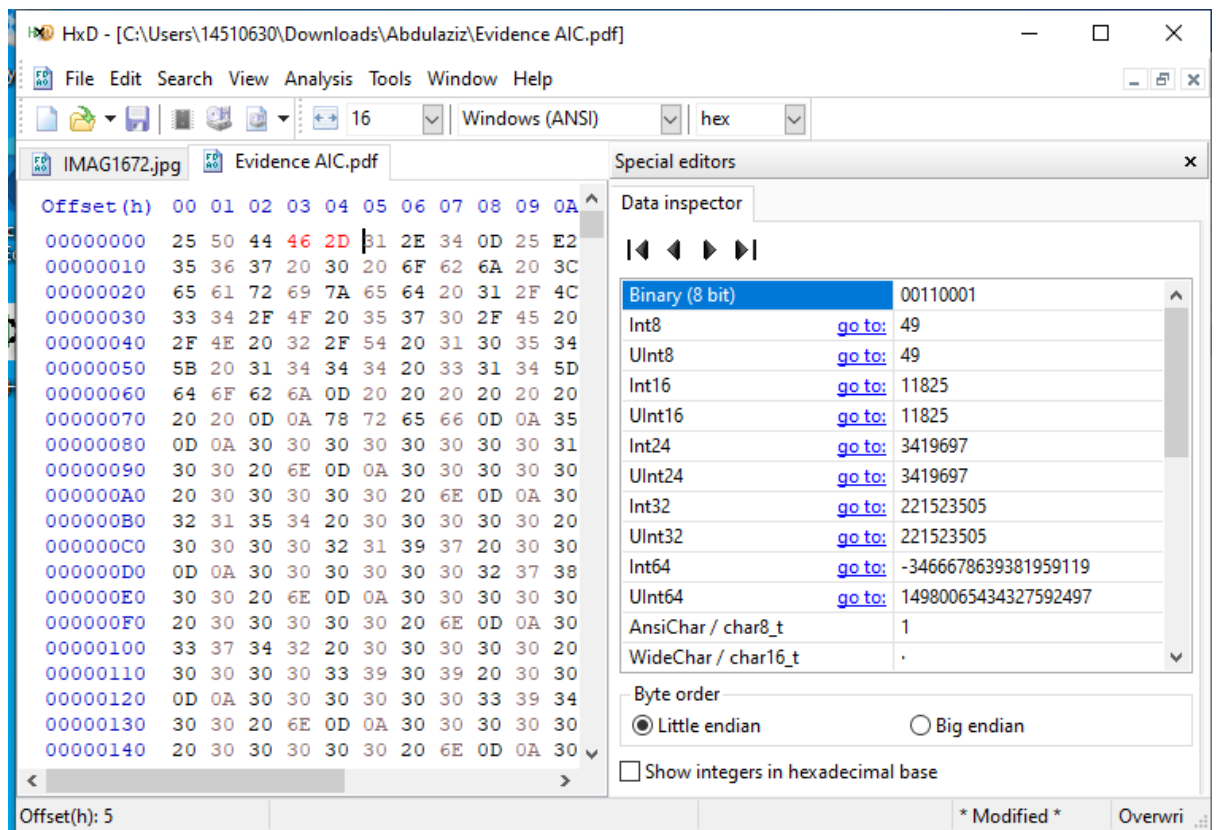
Examine the Exif photo. Where was it taken? (Nearest Suburb/City/State/Country.)

Balmoral, Mosman, Sydney, NSW Australia.

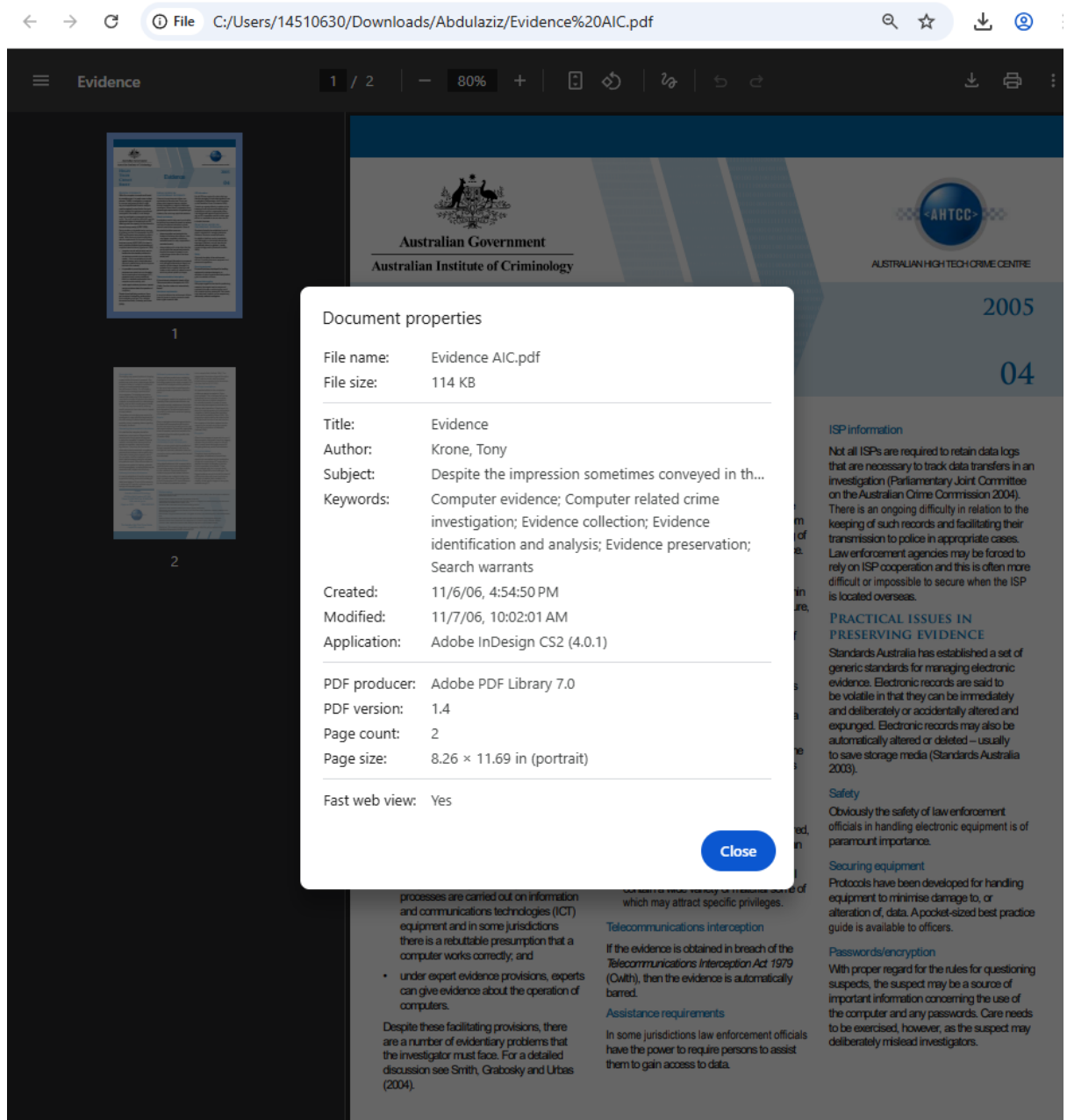
Explain how you located the device that took the photo? Found in the properties a GPS section so checked google for tools and used this tool pic2map.com

b) `abdulaziz@DESKTOP-Q8EFH82:/mnt/c/Users/14510630/Downloads/Abdulaziz$ xxd Evidence\ AIC.pdf | head -n1`
`00000000: 2550 4447 5f31 2e34 0d25 e2e3 cfd3 0d0a %PDG_1.4%. ....`

What is wrong with the HEX header? The beginning of the pdf file supposed to be %PDF-1.4 however, here as we can see it is %PDG-1.4



Also, take a screenshot of the Metadata showing:



What tool did you use? Google Chrome

c) Describe 25 e2 e3 cf d3. Is it ASCII?

It is the binary marker in a PDF. Not ASCII, because those bytes are Not normal characters.

They are there to tell the system the file has binary data.

What is the purpose of 25 e2 e3 cf d3 in a file?

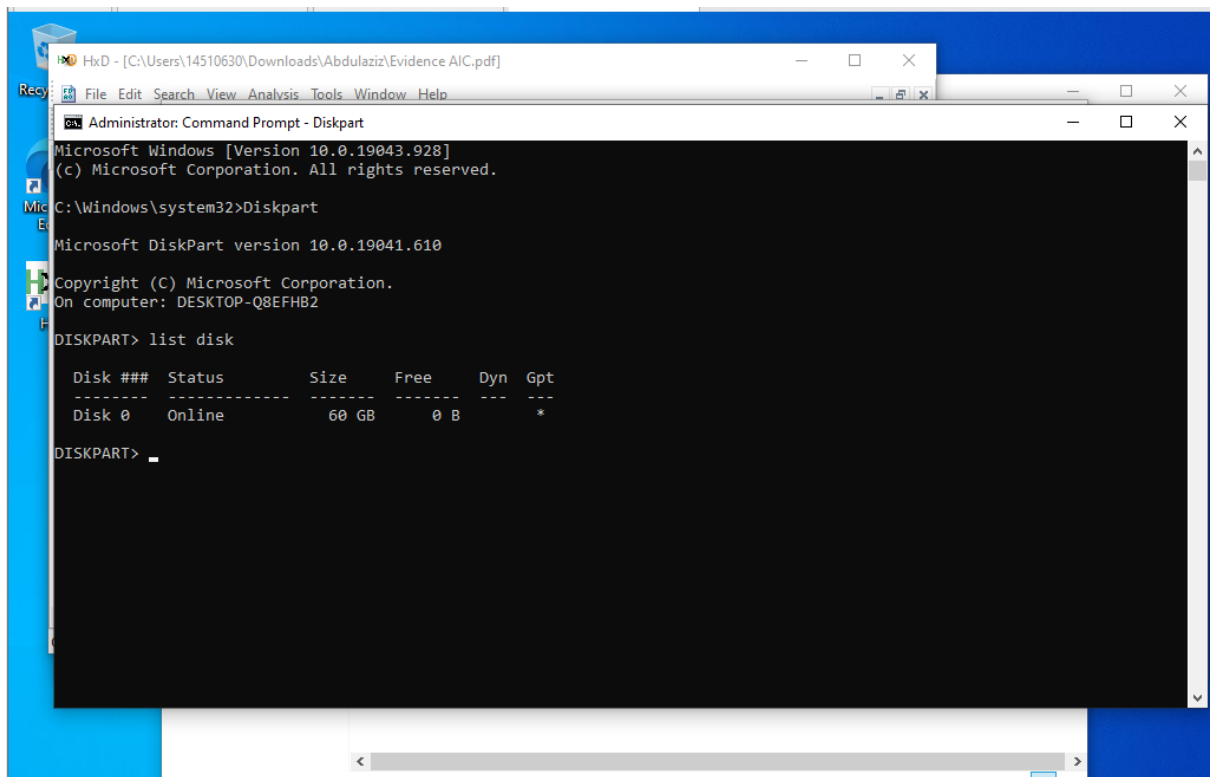
It help programs recognize the PDF as binary so it is read correctly.

Where will you find it?

At the start of a PDF file, right after the line %PDF-1.4 in the header

Q1 Week Disk Imaging Questions (Replace this with yours)

- a) What is a GUID? How can you tell the GUID version? A GUID is a unique ID number used to identify things like USBs or partitions. You can tell the version by looking 13th digit if it is 1, it's a Version 1 GUID.



The screenshot shows a Windows desktop with a taskbar at the bottom. A file explorer window titled 'HxD - [C:\Users\14510630\Downloads\Abdulaziz\Evidence AIC.pdf]' is open in the background. In the foreground, an 'Administrator: Command Prompt - Diskpart' window is open. The command prompt shows the following text:

```
Microsoft Windows [Version 10.0.19043.928]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>Diskpart

Microsoft DiskPart version 10.0.19041.610

Copyright (C) Microsoft Corporation.
On computer: DESKTOP-Q8EFHB2

DISKPART> list disk

Disk ###  Status         Size      Free      Dyn  Gpt
-----  -
Disk 0    Online         60 GB     0 B           *
```

The command prompt is currently at the 'DISKPART>' prompt.

c)

```

abdulaziz@DESKTOP-Q8EFHB2:/mnt/c/Users/14510630/Downloads/Abdulaziz$ fsstat C08InChp.dd | grep -m30 .
FILE SYSTEM INFORMATION
-----
File System Type: FAT12
OEM Name: MSDOS5.0
Volume ID: 0x94de152b
Volume Label (Boot Sector): NO NAME
Volume Label (Root Directory): EMTS
File System Type Label: FAT12
Sectors before file system: 0
File System Layout (in sectors)
Total Range: 0 - 2879
* Reserved: 0 - 0
** Boot Sector: 0
* FAT 0: 1 - 9
* FAT 1: 10 - 18
* Data Area: 19 - 2879
** Root Directory: 19 - 32
** Cluster Area: 33 - 2879
METADATA INFORMATION
-----
Range: 2 - 45782
Root Directory: 2
CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 512
Total Cluster Range: 2 - 2848
FAT CONTENTS (in sectors)
-----
33-33 (1) -> EOF
abdulaziz@DESKTOP-Q8EFHB2:/mnt/c/Users/14510630/Downloads/Abdulaziz$

```

What is the Volume Label (Root)? EMTS

How big are the two FAT tables? (how many sectors) each one is 9 sectors long.

Are they both the same? Why? Yes they're exactly same, made for backup, so if one gets damaged, the other can be used to recover it

Use fls on C08InChp.dd What option lets you see inside the folders? -r

```

abdulaziz@DESKTOP-Q8EFHB2:/mnt/c/Users/14510630/Downloads/Abdulaziz$ fls -r C08InChp.dd
d/d 12: Homework
+ r/r * 262: Odessey02.txt
+ r/r * 264: Odessey03.txt
+ r/r * 266: Odessey04.txt
+ r/r * 268: Odessey05.txt
+ r/r * 270: Odessey06.txt
+ r/r * 272: Odessey07.txt
+ r/r * 274: Odessey08.txt
+ r/r * 980: Odessey09.txt
+ r/r * 982: Odessey10.txt
+ r/r * 984: Odessey11.txt
+ r/r * 986: Odessey12.txt
+ r/r * 988: Odessey13.txt
+ r/r * 990: Odessey14.txt
+ r/r * 992: Odessey15.txt
+ r/r * 994: Odessey16.txt
+ r/r * 996: Odessey17.txt
+ r/r * 998: Odessey18.txt
+ r/r * 1000: Odessey19.txt
+ r/r * 1002: Odessey20.txt
+ r/r * 1004: Odessey21.txt
+ r/r * 1006: Odessey22.txt
r/r 3: EMTS (Volume Label Entry)
d/d * 5: New Folder
d/d 8: Vacation Pictures
+ r/r * 230: gametour2.exe

```


Name three deleted files (ones that have their names intact) and give their folders and inodes.

Folder: Homework

262: Odessey02.txt

264: Odessey03.txt

266: Odessey04.txt